Introduction to Computer Simulation
Home Work 8

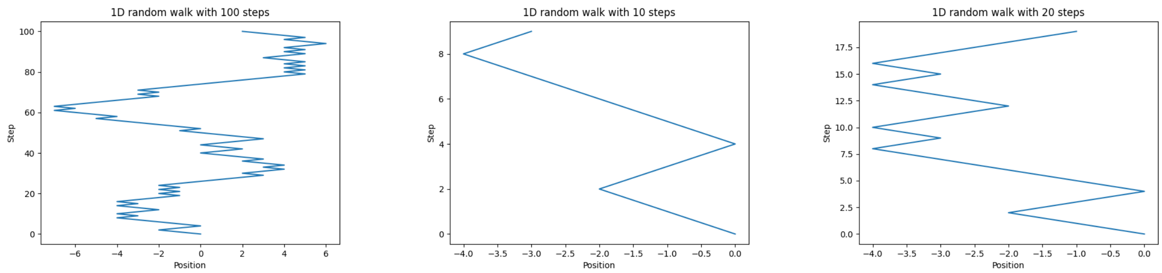
Guilherme Schewtschik (3748052)

December 17, 2025

Question 1

1.a)

For this question the probability of going right is set to the same as for going left,p lotting the graphs for the random walks we get:



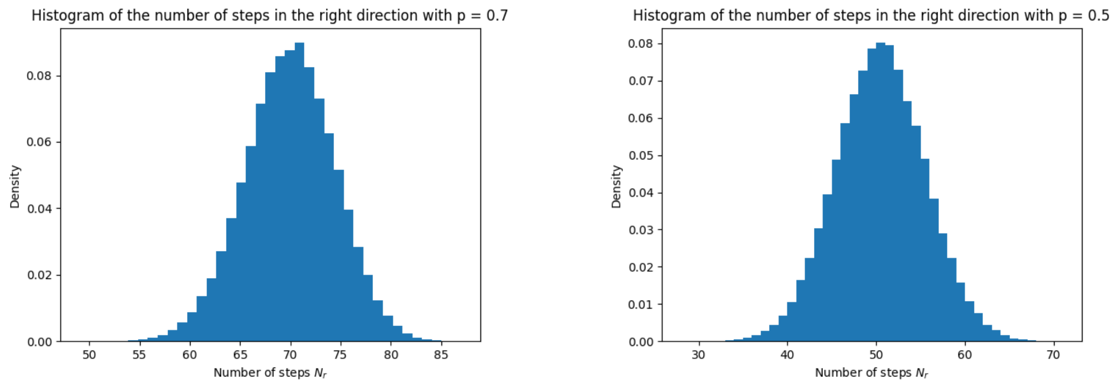
(a) $N = 100$ (b) $N = 10$ (c) $N = 20$

Figure 1: Random walks for different N

We can see that for higher values of N leads to the possibility of greater divergence from the initial position $x_0 = 0$

1.b)

We can plot a histogram for the number of right steps N_r of the random walks:

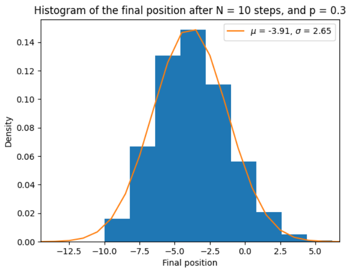


(a) $N = 100$ (b) $N = 100$

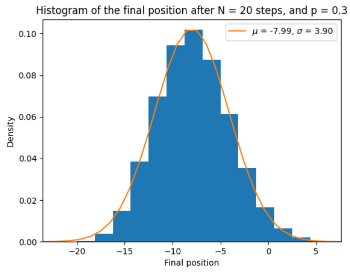
We can see that for $N = 100$ and $p = 0.5$ the most probable result is around 50 i.e. half of all steps, while for $p = 0.7$ it hovers around 70, it is clear that for a smaller number of steps the mean will remain the same, only having a smaller standard deviation.

1.c)

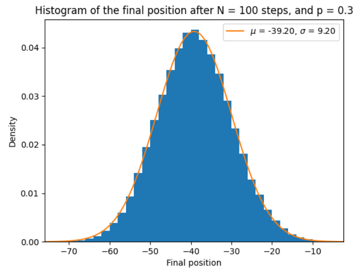
We can see that for smaller N the resolution of the histograms is lower, and as discussed in the previous question a smaller standard deviation, the simulation breaks at $N = 100$, $p = 0.9$.



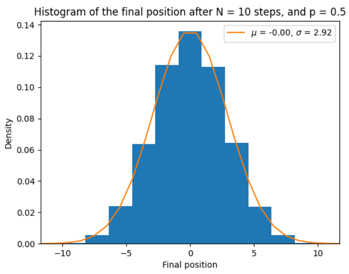
(a) (a)



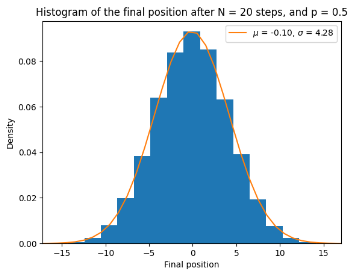
(b) (b)



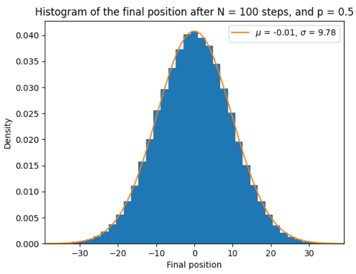
(c) (c)



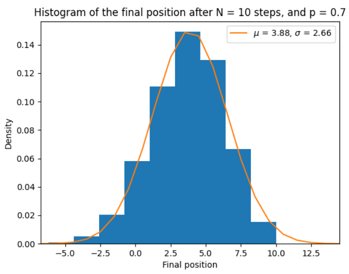
(d) (a)



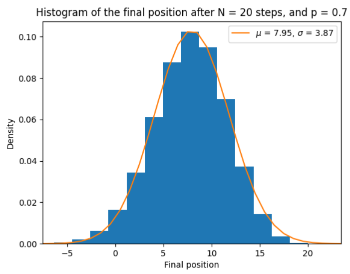
(e) (b)



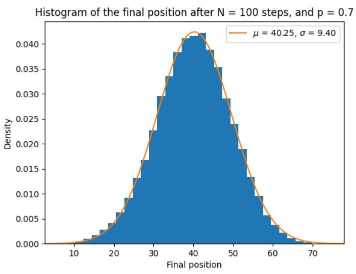
(f) (c)



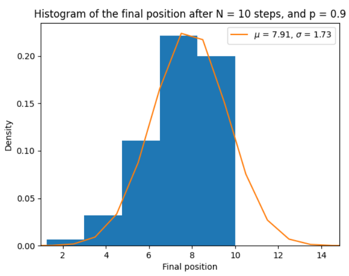
(g) (a)



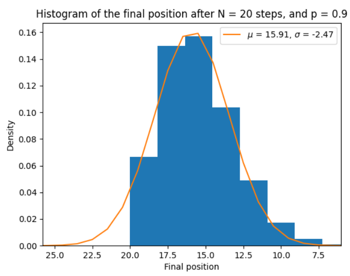
(h) (b)



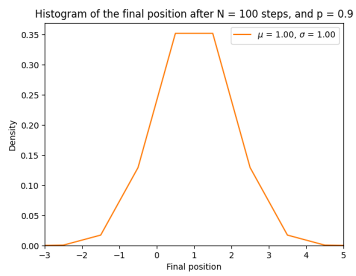
(i) (c)



(j) (a)



(k) (b)

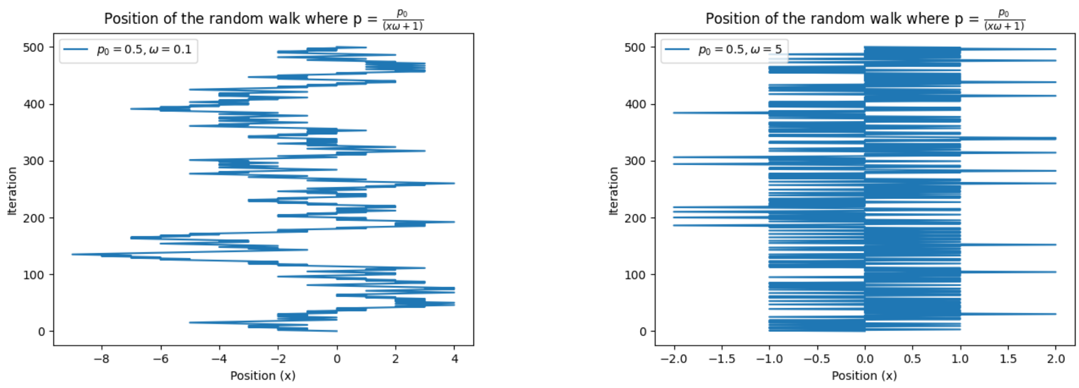


(l) (c)

Figure 3: Twelve graphs arranged in a 3×4 grid.

1.d)

Taking the probability as a function of the position such as in a oscillator, i.e. the further the particle is the stronger the probability of it returning into the initial position we arrive at the following graphs:



(a) $N = 500$

(b) $N = 500$

we can see that for a high ω the deviation from the center position is small in comparison to the small ω value.

Question 2

The self-avoiding walks take place in a two-dimensional square lattice up to $N = 20$. The walk starts at the $\vec{r} = (0, 0)$ and at each step may move to one of the four nearest-neighbor lattice sites, if the point has not been visited before. For each N we compute the total number of walks $Z(N)$ and $\langle R_{ee}^2 \rangle$.

For large N, the mean squared end-to-end distance is expected to obey the scaling law

$$\langle R_{ee}^2 \rangle \sim N^{2\nu}, \tag{1}$$

with the Flory exponent $\nu = 3/4$ in two dimensions. Plotting $\ln\langle R_{ee}^2 \rangle$ vs $\ln N$. We can see that the data follow a straight line for Smaller N, in good agreement with the expected slope $2\nu = 3/2$. Deviations at big N are due to finite-size effects.

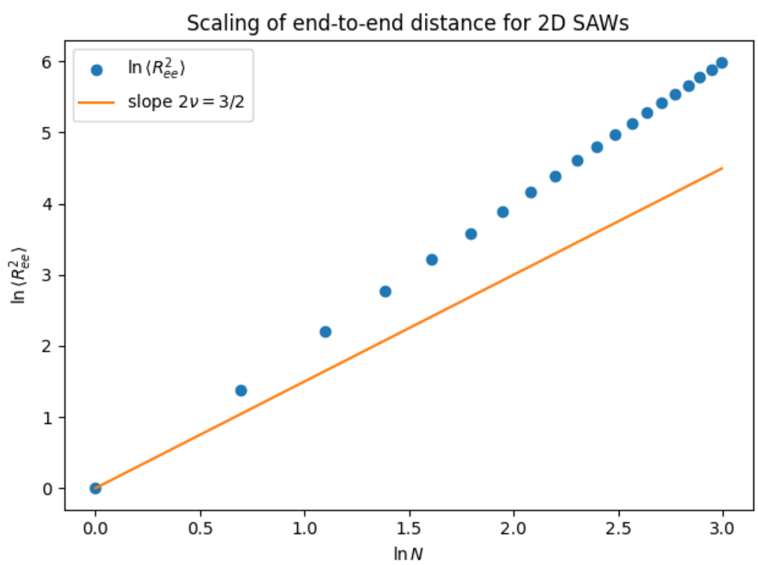


Figure 5: Log–log plot of $\langle R_{ee}^2 \rangle$ as a function of N for two-dimensional self-avoiding walks. The yellow line is the expected scaling with exponent $\nu = 3/4$.

The total number of self-avoiding walks grows rapidly with N and is expected to scale asymptotically as

$$Z(N) \sim \mu^N N^{\gamma-1}, \tag{2}$$

where μ is the connective constant of the square lattice and $\gamma = 43/32$ in two dimensions. Figure 6 shows a log–log plot of $Z(N)$ as a function of N. Although the dominant behavior is exponential, the plot illustrates finite-size effects and subleading corrections for small N.

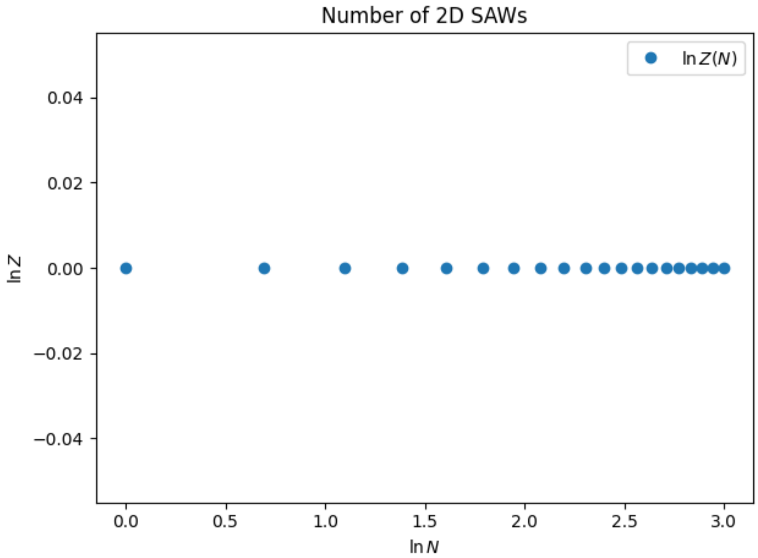


Figure 6: Log–log plot of the total number of self-avoiding walks $Z(N)$ as a function of the number of steps N.